

Math 421: Homework 6

Solutions

Due: Oct 24, 2025

Problem 1: (Ross 9.3) - Suppose $\lim a_n = a$ and $\lim b_n = b$. By Theorem 9.4, $\lim a_n^2 = \lim a_n \cdot a_n = \lim a_n \cdot \lim a_n = a \cdot a = a^2$. Using Theorem 9.4 again, $\lim a_n^3 = \lim a_n \cdot a_n^2 = \lim a_n \cdot \lim a_n^2 = a \cdot a^2 = a^3$. By Theorem 9.2, $\lim 4a_n = 4 \lim a_n = 4a$. Using Theorem 9.3, $\lim a_n^3 + 4a_n = \lim a_n^3 + \lim 4a_n = a^3 + 4a$. Using Theorem 9.4, $\lim b_n^2 = \lim b_n \cdot b_n = \lim b_n \cdot \lim b_n = b \cdot b = b^2$. Using Theorem 9.3, $\lim b_n^2 + 1 = \lim b_n^2 + \lim 1 = b^2 + 1$. Since $b^2 + 1 \neq 0$ and $b_n^2 + 1 \neq 0$ for all n , we can apply Theorem 9.6 and get

$$\lim s_n = \lim \frac{a_n^3 + 4a_n}{b_n^2 + 1} = \frac{\lim a_n^3 + 4a_n}{\lim b_n^2 + 1} = \frac{a^3 + 4a}{b^2 + 1}.$$

Problem 2: (Ross 9.9 b)

Theorem. Let s_n and t_n be sequences such that there exists an N_0 such that $s_n \leq t_n$ for all $n > N_0$. If $\lim t_n = -\infty$, then $\lim s_n = -\infty$.

Proof. Let s_n and t_n be sequences such that there exists an N_0 such that $s_n \leq t_n$ for all $n > N_0$. Let $M < 0$. Since $\lim t_n = -\infty$ there exists $N \geq N_0$ such that $t_n < M$ for all $n > N$. Then $s_n \leq t_n < M$ for all $n > N$ and therefore $\lim s_n = -\infty$. $\square$

Problem 3: (Ross 9.10 b)

Theorem. $\lim s_n = +\infty$ if and only if $\lim(-s_n) = -\infty$.

Proof.

($\Rightarrow$) Let $M < 0$. Since $\lim s_n = +\infty$, there exists N such that for all $n > N$, $s_n > -M > 0$. This implies $-s_n < M$ for all $n > N$, so $\lim(-s_n) = -\infty$.

($\Leftarrow$) Let $M > 0$. Since $\lim(-s_n) = -\infty$, there exists N such that for all $n > N$, $-s_n < -M < 0$. This implies $s_n > M$ for all $n > N$, so $\lim(s_n) = +\infty$. $\square$

Problem 4: (Ross 10.2)

Theorem. All bounded decreasing sequences converge.

Proof. Let (s_n) be a bounded decreasing sequence. Let S denote the set $\{s_n : n \in \mathbb{N}\}$, and let $v = \inf S$. Since S is bounded and non-empty, by the Completeness Axiom, we know $v \in \mathbb{R}$. We will show $\lim s_n = v$. Let $\epsilon > 0$. Since $v + \epsilon$ is not a lower bound for S , there exists N such that $s_N < v + \epsilon$. Since (s_n) is decreasing, we have $s_N \geq s_n$ for all $n \geq N$. Since $s_n \geq v$ for all n , $n > N$ implies $v \leq s_n \leq v + \epsilon$, or $0 \leq s_n - v < \epsilon$, so that $|s_n - v| < \epsilon$. Therefore $\lim s_n = v$ and the sequence converges. $\square$

Problem 5: (Ross 10.5)

Theorem. If (s_n) is an unbounded decreasing sequence, then $\lim s_n = -\infty$.

Proof. Let (s_n) be an unbounded decreasing sequence. Let $M < 0$. Since the set $\{s_n : n \in \mathbb{N}\}$ is unbounded and it is bounded above by s_1 , it must be unbounded below. Hence, for some $N \in \mathbb{N}$, we have $s_N < M$. Thus, since our sequence is decreasing, $n > N$ implies $s_n \leq s_N < M$, so $\lim s_n = -\infty$. $\square$

Problem 6: (Ross 10.7)

Theorem. *If S is a bounded nonempty subset of $\mathbb{R}$ such that $\sup S$ is not in S , then there exists a sequence (s_n) of points in S such that $\lim s_n = \sup S$.*

Proof. Let S be a bounded nonempty subset of $\mathbb{R}$ such that $\sup S$ is not in S . Since $\frac{1}{n} > 0$ for all $n \in \mathbb{N}$, there exists $s_n \in S$ such that $\sup S - \frac{1}{n} < s_n < \sup S$. Since $\limsup S - \frac{1}{n} = \sup S = \limsup S$, by the squeeze lemma, $\lim s_n = \sup S$. $\square$